SOFTWARE ENGINEERING

LAB ASSIGNMENT 06

Sequence Diagram — Payment Processing

Digital Wallet Management System (DWMS)

SUBMITTED BY — TEAM 5

Suryansh Dwivedi — 2400290100314
Shubh Tiwari — 2400290100296
Subrat Dwivedi — 2400290100311
Tejasvi Gupta — 2400290100322

Department of Computer Science & Engineering

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Project basis: Team 5 Software Requirements Specification, Version 1.0

1. Introduction

Sequence diagrams document time-ordered interactions between external actors and internal system components. They make dependencies explicit and help designers check whether requests, validations, external API calls, persistent updates, and responses occur in a safe order.

1.1 Objectives

- Model the DWMS payment-processing scenario for a merchant or supported payee.
- Show validation and user confirmation before payment processing.
- Represent the external payment gateway response and final transaction status.

2. Selected Scenario: Merchant Payment

An authenticated customer enters payee details and an amount. The DWMS validates the payment request and available wallet balance, asks the customer to confirm, then processes the payment through the relevant gateway. The final success or failure is recorded and reported to the user.

2.1 Relevant SRS Requirements

- REQ-PAY-1: allow an authenticated user to initiate a payment to a supported payee.
- REQ-PAY-2: validate the request, including sufficient wallet balance.
- REQ-PAY-3: require explicit user confirmation before committing the payment.
- REQ-PAY-4: report the final success/failure status to the user.
- REQ-PAY-5: record every payment attempt, successful or failed, in the transaction ledger and audit log.

UML DIAGRAM

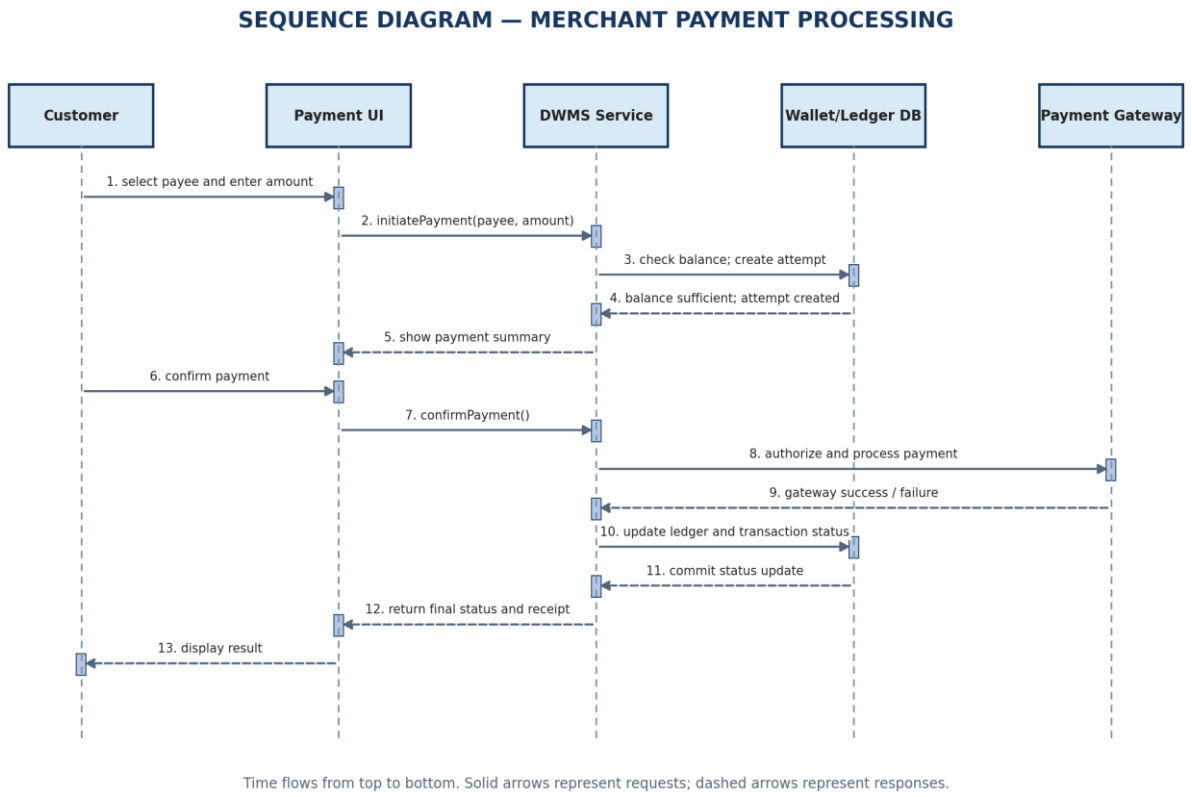


Figure 1. Sequence diagram for merchant payment processing.

3. Interaction Explanation

1. The customer selects a payee and enters the amount in the Payment UI.
2. The UI sends the payment request to the DWMS service.
3. The service checks the wallet balance and creates a transaction attempt.
4. After the customer confirms, the service sends an authorization/processing request to the payment gateway.
5. The gateway returns a status. The DWMS records the final transaction status and ledger/audit information.
6. The service returns the final status and receipt to the UI, which displays the result to the customer.

4. Design Considerations

- Do not treat a request as successful until a final gateway status is available.
- Failed payment attempts must also be recorded for traceability.
- External gateway details and vendor versions are unspecified in the SRS and should remain behind an interface.
- Network failures and delayed gateway responses should be handled without duplicating or corrupting financial records.

Conclusion

The payment sequence clarifies the boundary between DWMS responsibilities and the external gateway. It emphasizes balance validation, explicit confirmation, recording of both successful and failed attempts, and clear final feedback. These interactions provide a basis for integration tests and failure-handling tests.